Nocturnal Sleep in Separated Monkey Infants

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• **Nocturnal sleep was recorded from ten unrestrained, group-living *Macaca nemestrina* (pigtail) monkey infants, using implantable multichannel biotelemetry systems, during the agitation-depression behavioral reaction that follows maternal separation. Sleep disturbances during the four nights of separation were characterized by decreases in rapid eye movement (REM) time and in the number of REM periods, and increases in REM latency. Time awake and number of arousals were increased. Slow-wave sleep was not significantly affected.**

Sleep pattern changes were most pronounced the first separation night, and tended to decrease as separation continued, whereas behavioral measures of depression tended to increase as separation continued (up to four days). Sleep patterns returned to normal following reunion with the mother. Those infants who had the most severe sleep disturbances the first separation night (more time awake, less total sleep, less REM) also tended to become most depressed behaviorally later in the separation period.

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The premature loss of the mother is a severely stressful experience for the young of most higher-order animals. This is especially true among the primates; in human infants, loss of the mother (or lack of proper mothering) may result in the syndrome of "anaclitic depression",[1] and in older children to a "protest-despair" type of reaction.[2,3] In monkey infants, maternal separation also frequently results in a "protest-despair"[4] or "agitation-depression" behavioral reaction[5,6] that is quite similar to that of humans.[7] The depressive component of the agitation-depression reaction seen in separated monkey infants has been thought to represent a possible animal model of certain loss-related human depressive disorders.[5-9]

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In infant monkeys, the period of agitation that immediately follows maternal separation is characteristically accompanied by increases in vocalization and locomotive activity, often of a "searching" nature. This is followed by a period of behavioral "depression," with decreased locomotor behavior, slowness of movement, marked decreases in play behavior, and a slouched posture.[5,6] The depressive reaction varies in severity over a period of several days, and usually remits spontaneously and slowly, although occasionally a separated infant dies, with no apparent cause being found on autopsy (I. C. Kaufman, MD, and A. J. Stynes, MS, unpublished data). While the agitation or protest component of the reaction to separation has been observed in all species of monkeys studied to date,[5,6,10-17] the depressive component appears to be most prominent in pigtail (*Macaca nemestrina*) infants. The reasons for such apparent individual and species differences are not yet clear, although both Hinde and Spencer-Booth[18] and Kaufman[19] have implicated ontogenetic (experiential) mechanisms.

While this response is most interesting from a behavioral standpoint, our understanding of the mechanisms underlying this dramatic reaction to the premature disruption of an important attachment bond is hampered if we have access only to behavioral data. A more definitive understanding of the phenomena requires that we obtain data concerning the physiological status of the separated monkey infant as well; this is even more important if we are to propose this as an animal model of relevance to studies of human depressive disturbances.

Operationally, the situation is complicated by the fact that our experimental paradigm for studying the effects of maternal separation requires that the infants be raised by their natural mothers in social groups, and that they not be restrained, isolated, or otherwise artificially influenced. This precludes the conventional hard-wire methods of recording physiological data. Accordingly, during the last several years, our laboratory has been developing the

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implantable biotelemetry technology necessary to monitor multivariable physiological data simultaneously with behavior in the totally unrestrained monkey infant living in its social group.[20]

In a previous report on the first 36 hours of maternal separation in four pigtail infants, we described a marked disturbance in sleep patterns on the first separation night, with a decrease in total sleep time, and a large decrease in rapid eye movement (REM) sleep.[9] Sleep would appear to be an important variable to study in this animal model, as it is often significantly altered in the affective disorders in man,[21,22] and changes in certain sleep variables have been proposed as physiological markers that might permit a more meaningful grouping of affective disorders.[21,23] In addition, considerable work is in progress on the relationship of specific neuroanatomical and neurochemical systems to sleep regulation, which may ultimately relate to neuroanatomical and/or neurochemical mechanisms underlying the affective disorders.

This article reports changes in nocturnal sleep patterns accompanying four days of maternal separation in ten pigtail monkey infants as well as the subsequent changes occurring after reunion with the mother. We will demonstrate that there are pronounced and specific alterations in sleep patterns accompanying the agitation-depression reaction. We believe that such information will improve our understanding of the organismic nature of the reaction to separation, which plays such a prominent role in psychiatry, and will also provide us with important additional information on how this animal model system may be relevant to studies of human affective disorders.

METHODS

All ten infants (five males and five females) were naturally born in the Department of Psychiatry Primate Laboratory. They were raised by their mothers in social groups containing eight to ten adult females, some with offspring, and one adult male. The groups were housed in pens with glazed cinderblock walls containing one-way glass observation windows, wire mesh ceilings, and shelving and pipes to provide sitting and climbing space. The groups were on a 13:11-hour light-dark cycle, with timer controlled lights-off between 8 PM and 7 AM. Pen temperature was maintained at a nominal 25 C. Feeding took place at 9 AM daily, and water was available ad lib.

A Silastic-encapsulated multichannel biotelemetry transmitter was surgically implanted in the infants when they were about 23 weeks of age (range, 20 to 33 weeks). Designed by us, the telemetry unit is a seven-channel pulse-amplitude modulated FM system that transmits electro-oculogram, electromyogram, three channels of EEG (from frontal, centrotemporal, and parieto-occipital regions), as well as heart rate and body temperature.[24] The telemetry electronics and power supply were implanted in the anterior abdominal walls, and the electrode leads were tunneled subcutaneously to their respective recording sites.[25] Immediately following recovery from surgical anesthesia, the infants were returned to their mothers. Sutures were removed ten days postoperatively, by which time the infants' behavior had returned to normal. The implanted telemetry systems did not interfere with the infants' individual and social behavior, and since there were no external components, the implanted animals were indistinguishable from other infants in the group.

Physiological data recording began between 12 days and four weeks postoperatively. Our overall system for recording and analyzing physiological and behavioral data in monkey infants has been described previously.[26] Briefly, the telemetered signal was received on a modified commerical FM receiver, demodulated, and the physiological signals recorded on paper and magnetic tape. Time synchronization was provided by a time code generator that produced an IRIG B time code for the tape and a simultaneous slow code for the paper.

In nine of the ten infants, quantified behavioral data were also collected (one infant was not systematically observed). Six five-minute behavioral observation sessions were scheduled daily. A trained observer watched the animals through the one-way windows and dictated observed behavior onto an audio cassette. The audio cassette was later scored on a clock-counter device, or directly on the laboratory computer providing duration and frequency measurements of all behaviors, with preservation of temporal sequences and absolute time of occurrence.[26,27] Behavioral observation sessions for a given day or other appropriate period were grouped together, and quantified scores for the behaviors of interest were calculated on the basis of the percentage of total time observed. We will present quantified data on four specific behaviors: (1) time in ventral-ventral contact with the mother (VVcon); (2) play behaviors, either alone or with other infants (Play); (3) locomotion behaviors including walking, running, and climbing (Motion); and (4) the slouched posture seen during depression (Slouch). These behaviors have been previously determined to be sensitive indicators of the agitation-depression reaction.[13] Wilcoxon signed-rank tests were performed on the four behavioral categories across animals for each experimental day, using the mean of the last three baseline days as the norm.

Following the collection of three to five consecutive days and nights of baseline physiological and behavioral data, the mother was separated from the infant. The infant remained in the social group in which it had been raised; the mother was housed separately, out of sight and sound of the infant, during the separation period. Following at least four days and nights of separation (several infants were separated for slightly longer periods), the mother was returned to the social group, and three to five consecutive days and nights of reunion data were recorded.

The all-night paper sleep records were hand-scored on a teletype terminal with paper tape punch. A specific teletype key corresponding to the stage of sleep was struck for each page of the paper record (20-second epoch), and the punched paper tape so generated was read by the computer, which provided a statistical summary of the night's sleep, a sleep stage plot, and digital tape storage for future analysis.[28] A list of the sleep variables scored each night for each infant is presented in Table 1. Sleep stage criteria were similar to those for adult *M nemestrina* monkeys,[29] with modifications appropriate for the younger animals.[30]

Nocturnal sleep in normal infant *M nemestrina* monkeys exhibits a considerable degree of night-to-night intraindividual variability seemingly unrelated to various measures of the preceding day's behavior, and in the absence of major environmental changes. However, when sleep variables are averaged over several nights, animals are quite similar to each other.[31] For the purpose of this report, we have grouped the sleep variable values for the ten infants.

For each sleep variable, a one-way ANOVA was performed across all animals by night.[32] In addition, a critical range for comparison of means was computed using Scheffé's method for significant deviations from baseline values.[33] Since the number of baseline nights varied from three to five in different infants, we used the sleep values obtained during the last three baseline nights for determining baseline mean values.

RESULTS

In all cases the mothers were separated from the infants and removed from the group in the early afternoon. The

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behavioral changes occurring in the nine infants on whom behavioral data were collected are illustrated in Fig 1. Eight of the nine infants exhibited the agitation reaction in response to separation, with increased motor activity, often of a "searching" nature (note increased motion behavior on day S, in Fig 1, which corresponded to the agitation reaction). Frequent "cooing" (distress vocalization) was also noted. The agitated behavior usually persisted until lights-out that night, at which time our behavioral observations ceased. The depressive component of the agitation-depression reaction usually began (was first noted by observers) the morning after separation (day S_2—the morning following the S_1 night). The infants were observed to be moving more slowly than normal, the amount of play behavior was diminished (Fig 1), and they would not respond to play initiation overtures made by other infants. The following days (the third and fourth days of separation) were usually accompanied by the most marked depressive behaviors, including the characteristic slouched posture.

Reunion with the mother was usually accompanied by an

Table 1.—Sleep Variables Calculated for Each Night's Sleep in Each Infant Monkey

Variable	Comment
Sleep latency	Time from lights-out to sleep onset
No. of arousals	All arousals more than 10 sec in duration
Total sleep time	Not including time awake during night
Awake	Time spent awake after sleep onset and before final awakening in the morning
Drowsy	Criteria of Reite et al[30]
Stage 2	Criteria of Reite et al[30]
Stage 3-4	Criteria of Reite et al[30] (stage 3 and 4 sleep are not separated)
REM	Criteria of Reite et al[30]
No. of REM periods	Criteria of Reite et al[31]
REM latency	Time from sleep onset to onset of first REM period
Inter-REM interval	Interval between starting times of successive REM periods[31]
Slow-wave sleep	Includes drowsy, stage 2, and stage 3-4 sleep

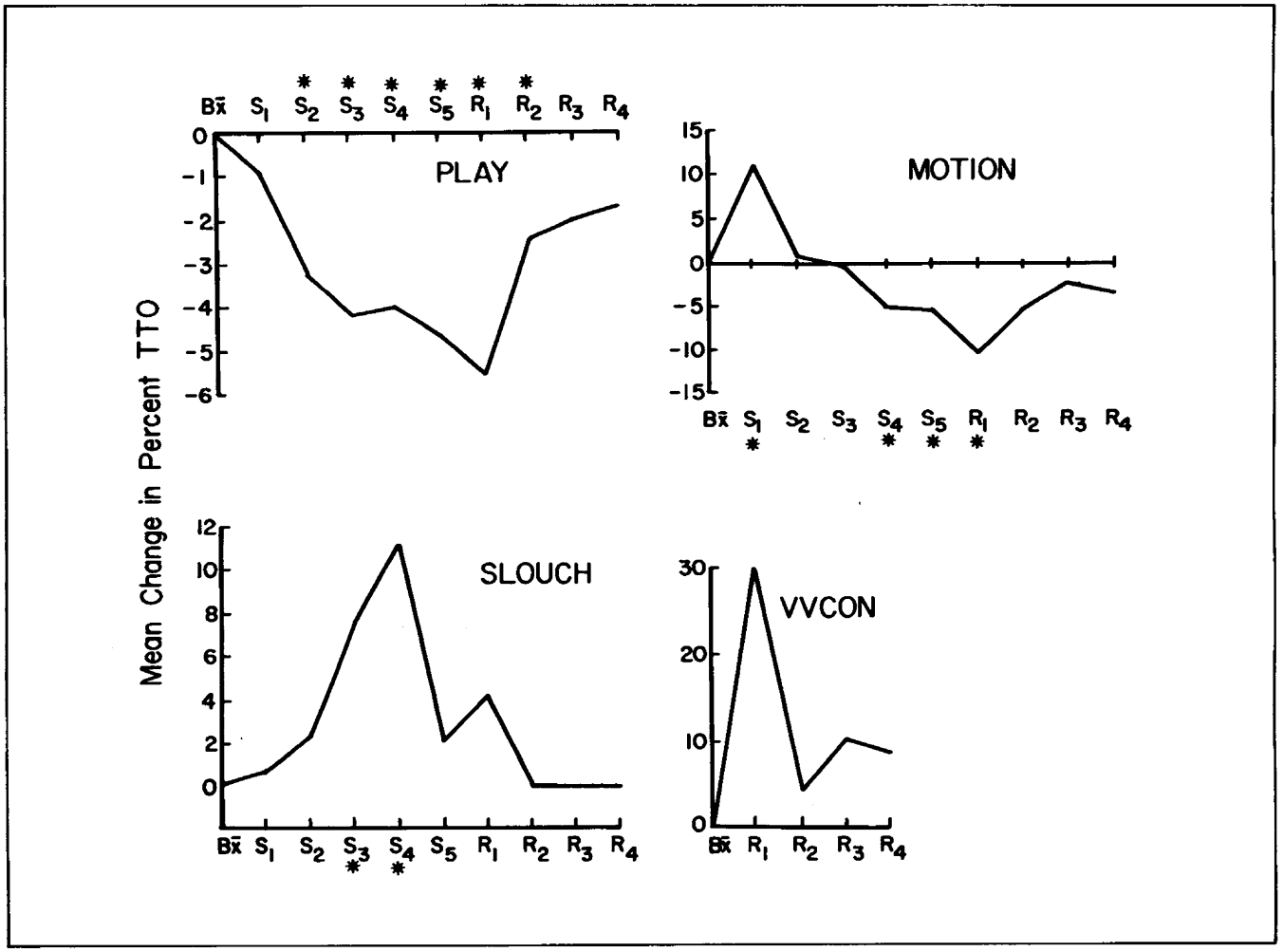


Fig 1.—Mean change (in percent) from baseline in total time observed (TTO) for four behaviors across animals. Asterisks denote those experimental days for which Wilcoxon signed-rank test indicated a significant change in behavioral categories across animals. S_1 through S_4 are postseparation days; R_1 through R_4 are postreunion days.

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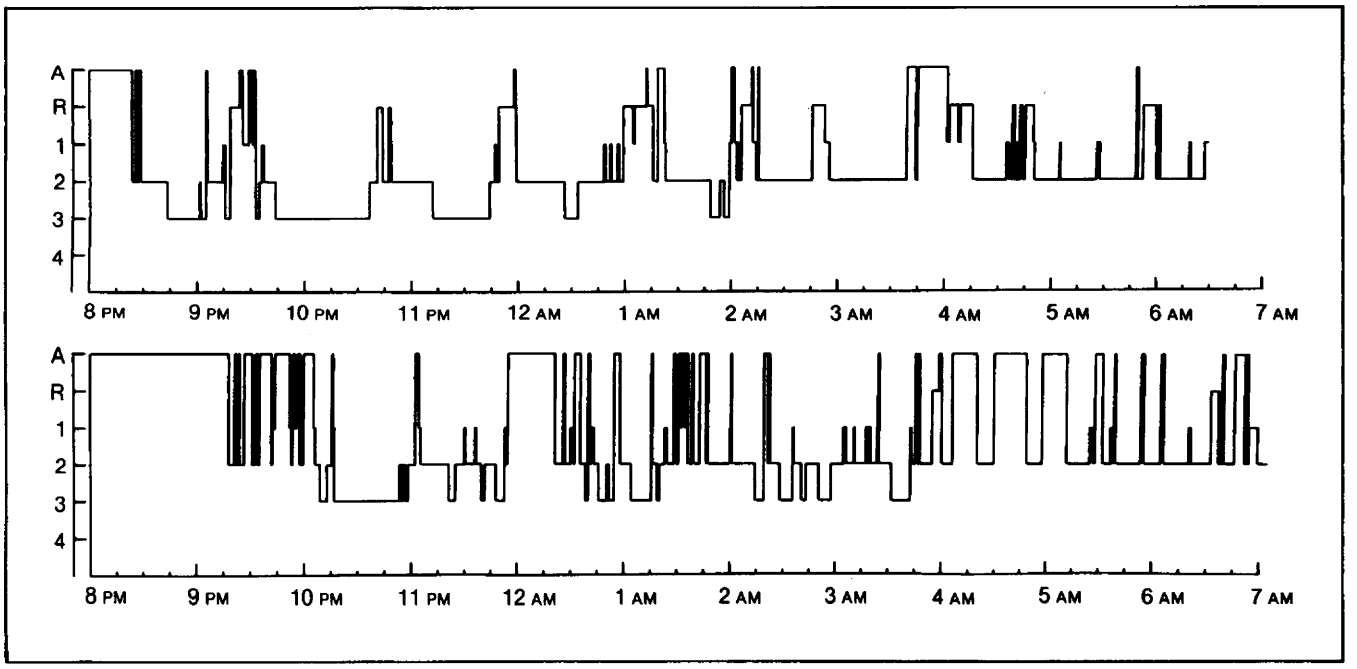


Fig 2.—All-night sleep plots from 26-week-old infant monkey during baseline night (top) and first separation (S_1) night (bottom). Time (X axis) is from 8 PM (lights-out) to 7 AM (lights-on). A indicates awake; R, REM; 1, drowsy; 2, stage 2; 3, stage 3-4 (combined).

Table 2.—Mean Sleep Values for Ten Infant Monkeys*

		Following Maternal Separation				Following Maternal Reunion				ANOVA	
	Baseline†	S_1	S_2	S_3	S_4	R_1	R_2	R_3	R_4	*F*	*P*
No. of nights	30	10	10	10	10	8	8	7	6	...	...
Sleep latency	35(3.4)	77‡	37	31	39	22	26	42	30	4.771	< .001
Total sleep	563(6.8)	471	507	524	536	582	578	562	580	6.34	< .001
Awake	51(5.2)	95	94	85	66	42	46	45	58	2.463	< .02
No. of arousals	15(2.4)	33‡	27	21	21	14	15	16	16	4.354	< .001
Drowsy	20(2.1)	27	22	27	26	26	33	29	28	.737	NS
Stage 2	297(5.5)	269	295	270	294	297	286	299	269	.781	NS
Stage 3-4	151(4.9)	146	146	169	161	164	159	141	170	1.054	NS
Slow-wave sleep	469(4.6)	442	463	466	484	487	478	469	467	1.090	NS
REM latency	64(6.4)	212‡	163	197‡	119	102	80	77	52	4.951	< .001
REM	90(4.6)	21‡	36‡	48‡	46‡	88	95	89	101	18.272	< .001
No. of REM periods	9.7(.29)	3.6‡	5.1‡	6.2‡	7.2	8.4	10.6	9.9	10.0	10.86	< .001
Inter-REM interval	58(1.3)	111	107	83	93	60	55	60	48	1.744	NS

*Data were not obtained for all infants for all reunion nights. Computer round-off error accounts for the fact that not all columns total correctly.
†Numbers in parentheses are standard error of the mean.
‡Significantly different from baseline by the Scheffé test.

intensification of the mother-infant relationship as measured by increased time spent in contact with the mother (VVcon) and disappearance of slouched behavior and cooing. Play behavior remained lower than during baseline levels, but had begun to recover by the second reunion day.

These behavioral changes during separation and reunion were similar to those previously described as characteristic of the agitation-depression reaction.[13]

Table 2 lists all sleep variables for the three baseline nights (grouped), and for individual separation (S_1 through S_4) and reunion (R_1 through R_4) nights. Those variables showing significant changes on the ANOVA were tested night by night using the Scheffé test for multiple comparisons to determine which of the separation or reunion nights differed significantly from baseline.

During the period shortly following lights-out on the first separation night, the separated infant could be heard "cooing" in the group, and group activity, as judged by sound level, was greater than normal. All infants had increased sleep latency the S_1 night ($P < .05$, Scheffé); the total number of arousals the S_1 night was markedly increased ($P < .05$, Scheffé), and the total amount of time spent awake during the night was increased (95 vs 51 minutes for baseline, S_1 not separable by Scheffé, but $F = 2.46$, $df = 10/88$, $P < .02$). Total sleep time (TST) was significantly decreased the S_1 night ($P < .05$, Scheffé). There was a marked increase in REM latency the S_1 night

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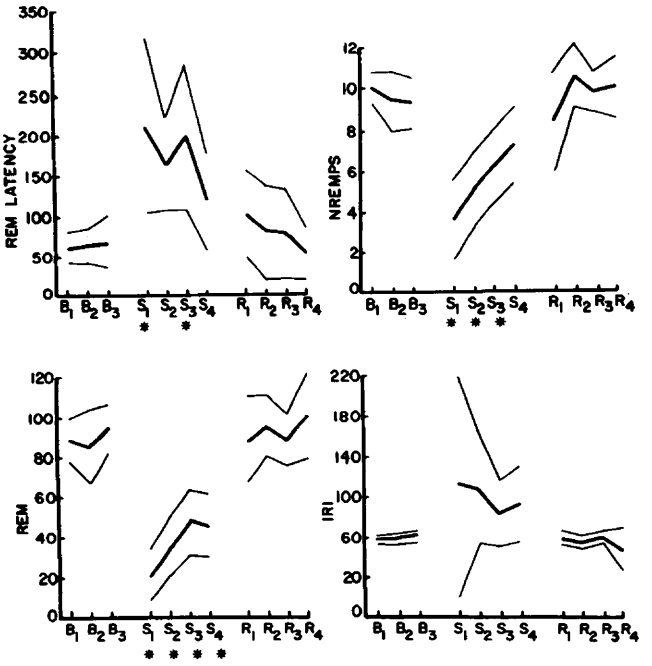


Fig 3.—Group mean values for REM latency (top left), total REM sleep (bottom left), number of REM periods (NREMPs, top right), and inter-REM interval (IRI, bottom right) for three baseline (B_1 through B_3), four separation (S_1 through S_4), and four reunion (R_1 through R_4) nights. Night-by-night mean values are bracketed by 95% confidence intervals for mean. Asterisks indicate significant difference from baseline ($P < .05$ or better), using Scheffé test. All values are in minutes.

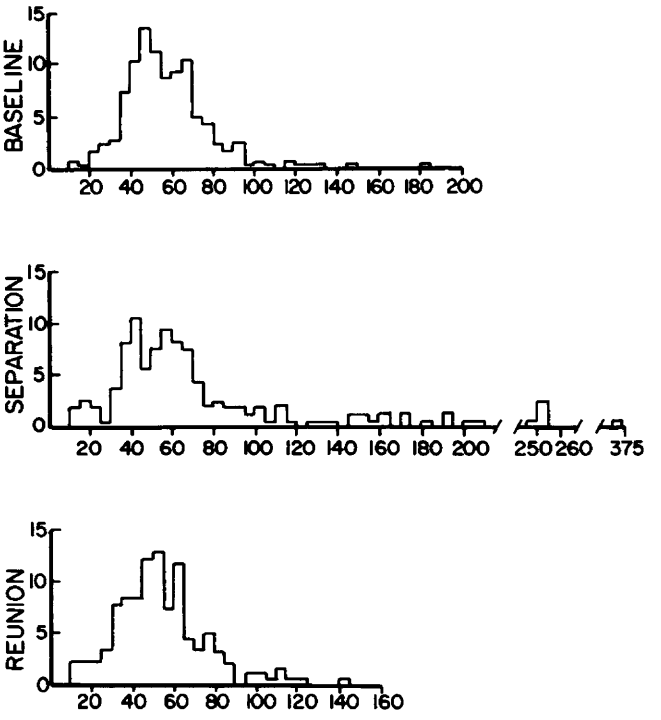


Fig 4.—Interval histograms of all inter-REM intervals during baseline, separation, and reunion in ten infant monkeys. Because of varying number of histograms were normalized by conversion to percent of total (Y axis). Time in minutes is plotted on X axis.

($P < .05$, Scheffé), and total REM time was greatly diminished ($P < .05$, Scheffé). Two of the infants had no REM sleep at all the S_1 night. The total amount of slow-wave sleep (SWS) did not differ significantly from the baseline mean, nor did the stage 3-4 contribution to total SWS. Figure 2, which includes a typical all-night sleep plot for the first separation night in one of the infants with a baseline night from the same infant for comparison, illustrates the changes described above.

The number of arousals remained increased throughout the separation period, returning to baseline following reunion with the mother (Table 2). Sleep latency, which was significantly increased the first separation night, was not significantly different from baseline during the remaining separation or reunion nights. Both the amount and the variability of time awake during the separation nights were increased, while mean time awake during reunion was unchanged from baseline. The TST was less than baseline values for all four separation nights (although only the S_1 night was separable with the Scheffé test), and was slightly (but not significantly) increased during reunion.

Although the variability of the measures of SWS, stage 2 sleep, and stage 3-4 sleep tended to increase during separation (and during reunion), the mean values did not differ significantly from baseline.

Figure 3 illustrates the four variables characterizing REM sleep. The REM latency (Fig 3, top left) was elevated throughout the separation period, with the Scheffé test separating the S_1 and S_3 nights as significantly different from baseline. Both the total amount of REM sleep and the number of REM periods remained decreased during the separation period (Fig 3, bottom left and top right), with the number of REM periods tending to return to normal over the four nights, but the amount of REM sleep taking a slight downturn on the fourth night of separation.

Both the mean and the variability of the inter-REM interval (IRI) were increased during separation (Fig 3, bottom right), although the changes were not statistically significant ($F = 1.74$, $df = 10/85$, NS). Three normalized histograms of all IRIs from all animals are illustrated in Fig 4. Modal values are seen to be quite similar during baseline, separation, and reunion. It would appear that the mean IRI during separation was influenced by the presence of a few very high values, which suggests that intervening REM periods were missed.

The general course of the behavioral changes following separation differed from the sleep pattern changes in that changes in Motion, Play, and Slouch behaviors tended to become more prominent with time (at least during the first four days of separation), whereas the sleep changes were most often greatest the first separation night, and tended to return toward normal as separation continued.

As a measure of the severity of the depressive reaction from a behavioral standpoint, we added together the change (from baseline) in Play and Motion behaviors on days S_4 and S_5 for individual infants, and called the resulting variable "D"—a quantitative measure of behav-

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ioral depression. This new "D" variable was positively correlated with the change (increase) in awake time the S_4 night ($r_s = 0.7833$, $P < .01$), and with the change (decrease) in REM time the S_4 night ($r_s = 0.7667$, $P < .05$), suggesting that those infants who were more severely depressed behaviorally also exhibited the greatest disturbance in sleep patterns, as measured by increased awake time and decreased REM time the S_4 night. Of even more interest is the apparent correlation between sleep pattern changes the first night of separation and behavioral changes later in separation. We found "D" to be positively correlated with the change in TST on the S_1 night ($r_s = .7667$, $P < .05$), with the change in awake time on both the S_1 and the S_2 nights ($r_s = .7167$, $P < .05$); $r_s = .7500$, $P < .05$), and with the change in REM on the S_1 night ($r_s = .5875$, $P < .06$, not statistically significant but strongly suggestive of a trend). Thus, those infants who had the most severe sleep disturbances the first night of separation tended to become more depressed behaviorally later in separation.

In all infants, reunion with the mother was accompanied by a return of REM time and number of REM periods to normal, with the R_1 night continuing to have a slightly (although not statistically significant) longer REM latency than baseline.

COMMENT

Our findings demonstrate that the agitation-depression behavioral reaction following maternal separation is accompanied by definite and specific effects on the infant's sleep patterns.

On the first night of separation, which follows the "agitation" reaction, it takes the infant longer to go to sleep, and sleep is interrupted by frequent arousals. The REM latency is markedly prolonged, and REM sleep is greatly diminished (or totally absent) this night. Slow-wave sleep, including deep sleep (stage 3-4) is not significantly altered in terms of total time, although it is more fragmented and tends to occur later, probably due in part to the increased sleep latency.

During the second and subsequent separation nights, the most prominent effects are the persistent increase in REM latency and the persistent diminution in total REM time and the number of REM periods. Sleep latency had returned to normal the second night of separation, and total SWS remained unchanged from baseline. Sleep was still somewhat more "restless" as judged by the increased number of arousals and the resulting slight increase in total time spent awake during the night. Our data are supportive of an effect whereby REM periods are replaced by arousals, suggesting an impairment in the ability to maintain the REM state.

Although the mean net loss of REM sleep during the four separation nights was 148 minutes, there was no evidence of a significant REM rebound, even after nights of reunion. Studies of total sleep deprivation in man and monkeys demonstrate that although stage 3-4 sleep recovers first, REM recovery is noted by the second night.[34-38] In a previous study, the selective deprivation of REM sleep in juvenile monkeys was found to be associated with a subsequent REM rebound largely complete in less than four days.[39] Thus, in the present study, if REM rebound or recovery were to occur, we would have expected to see it by the fourth reunion night, and so we conclude that it did not occur.

The sleep pattern changes we have described are quite similar to sleep changes constituting "first-night effect" in man. Human subjects, sleeping for the first time in a novel or strange (eg, laboratory) environment, have less REM time, increased REM latency, more time awake, and more frequent arousals, with little change in the total amount of slow wave (stage 3-4) sleep.[40-41] Infant monkeys normally sleep in contact with or enclosed by the mother. The absence of the mother at night is a unique experience for the infant, and represents a marked alteration in its normal sleep environment. Thus, a significant portion of the sleep changes we have seen in separated monkey infants may be related to the altered sleep environment.

Data from another experiment, however, militate against this being the sole explanation of our findings. In that study, to be reported elsewhere, we separated three infant pigtail monkeys from their mothers at birth, and raised them with cloth surrogate mothers. When at a later date their cloth mothers were removed from the cages, markedly and suddenly altering the infants' sleep environments, the infants exhibited disturbances in sleep patterns similar to those reported in this study, except that the disturbances were not as pronounced nor as long-lasting. We believe that the infant-surrogate attachment bond is considerably less intense than the infant-mother attachment bond, and that the sleep pattern changes accompanying its disruption are more likely related to the altered sleep environment per se rather than the loss of the mother as an important object or the disruption of a significant attachment bond. Extrapolating from these findings, it seems that the present findings may well be due in part to the altered sleep environment, but that other influences are likely operative as well, these accounting for the varying severity and persistence of the effects.

The loss of the mother is a complex stress, with elements biological, social, and quite likely psychological (eg, see Kaufman's discussion[42]). That these different elements contribute to varying degrees to the changes in sleep patterns seen during the reaction to separation is a reasonable assumption, but further work will be required to define the nature of the interaction.

Several studies have suggested that SWS is increased following exercise.[43-45] On this basis, we might have expected to see an increase in stage 3-4 sleep on the S_1 night, due to the increased Motion behavior accompanying the agitation reaction. The fact that we did not suggests that the increase in Motion may not be equivalent, in terms of physical exertion, to more strenuous exercise. This interpretation is supported by previous observations of normal (unseparated) pigtail infants, where we found that while more active infants tended to have more stage 3-4 sleep (averaged across several nights) than less active infants, there was no correlation between the amount of daytime activity and the amount of stage 3-4 sleep that night.[33] Also of importance, of course, is the fact that sleep on the S_1 night (and subsequent separation nights) is not analogous to sleep during recovery from a stress, for the

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stress of separation from the mother is a continuing one.

We believe the correlation between the magnitude of the sleep disturbances the first night of separation and the severity of the later behavioral depressive reaction is of considerable heuristic importance. Such a relationship, if confirmed in future experimental infants, suggests that it may be possible to predict the severity of the subsequent behavioral depressive reaction by evaluating the sleep pattern changes found immediately following separation. Clearly, in this case, we would wonder if a common mechanism underlies both types of changes, and whether such predictive capability would have more general applicability.

We consider the infant monkey separation reaction to be an internally consistent and reproducible paradigm in which both behavioral and physiological changes accompany a stress noted for producing similar behavioral reactions in young humans. The monkey case, being intrinsically somewhat less complex (eg, in terms of CNS and social structure), may permit us to study a nonhuman primate prototype of the human reaction to loss, or the disruption of an attachment bond, in a detailed and multivariable manner not possible in man.

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